Truth Table for $p \bar{\vee} q$

p	q	$p \bar{\vee} q$
T	T	F
T	F	T
F	T	T
F	F	F

Table 12.9

Example 12.16

Construct the truth table for $(p \bar{\vee} q) \wedge (p \bar{\vee} \neg q)$.

p	q	$\neg q$	r : $(p \bar{\vee} q)$	s: $(p \bar{\vee} \neg q)$	$r \wedge s$
T	T	F	F	T	F
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	F

Table 12.10

Also the above result can be proved without using truth tables. This proof will be provided after studying the logical equivalence.

12.3.3 Tautology, Contradiction, and Contingency

Definition 12.16

A statement is said to be a **tautology** if its truth value is always T irrespective of the truth values of its component statements. It is denoted by $\mathbb{T}$.

Definition 12.17

A statement is said to be a **contradiction** if its truth value is always F irrespective of the truth values of its component statements. It is denoted by $\mathbb{F}$.

Definition 12.18

A statement which is neither a tautology nor a contradiction is called **contingency**

Observations

1. For a tautology, all the entries in the column corresponding to the statement formula will contain T.
2. For a contradiction, all the entries in the column corresponding to the statement formula will contain F.
3. The negation of a tautology is a contradiction and the negation of a contradiction is a tautology.
4. The disjunction of a statement with its negation is a tautology and the conjunction of a statement with its negation is a contradiction. That is $p \vee \neg p$ is a **tautology** and $p \wedge \neg p$ is a **contradiction.** This can be easily seen by constructing their truth tables as given below.

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